

Reproduction of an Optimized Top-Hat and Homomorphic Filtering Method for Retinal Blood Vessel Segmentation

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Abstract

We reproduce the unsupervised retinal blood vessel segmentation method proposed by Ramos-Soto et al. [1], which combines an optimized morphological top-hat transform, homomorphic filtering, a two-dimensional matched filter, and a multilevel minimum cross-entropy thresholding algorithm optimized with the Harris Hawks Optimization (HHO) metaheuristic [2]. The pipeline separately segments thick and thin vessels, which are later merged. We implement the full method in Python using only NumPy, SciPy, Matplotlib and scikit-image, including our own implementation of HHO, and evaluate it on the training subset of the DRIVE dataset [3]. Our implementation reaches an average Dice coefficient of 0.451 ± 0.086 , with lower sensitivity than the values reported in the original article, mostly due to two binarization details that are not fully specified in the paper.

Keywords: retinal vessel segmentation, mathematical morphology, homomorphic filtering, Harris Hawks Optimization, DRIVE dataset.

1 Introduction

Retinal fundus imaging is a non-invasive way to directly observe blood vessels in the human body, and changes in their shape or width are linked to conditions such as diabetic retinopathy, hypertension and arteriosclerosis [1]. Manual segmentation of these vessels is slow and depends heavily on who draws it: two different observers rarely agree pixel by pixel. That is exactly the kind of problem automatic methods are meant to fix.

The article we reproduce, Ramos-Soto et al. [1], proposes an unsupervised pipeline that does not require a labeled training set, unlike deep-learning approaches. The method splits the problem into two branches, one tuned for thick vessels and one for thin vessels, since these differ substantially in width and contrast. Each branch relies on classical filtering (morphological and frequency-domain), while the thin branch additionally uses a metaheuristic optimization step (MCET-HHO) to select the segmentation threshold automatically.

We selected this article because its methodology overlaps strongly with the tools covered in this course (Fourier-domain filtering, mathematical morphology, image binarization), while also requiring us to learn and implement, from scratch, a technique that is not part of the syllabus: the Harris Hawks Optimization metaheuristic.

2 Methodology

2.1 Pre-processing

The green channel is extracted from the RGB fundus image, since it shows the highest vessel-background contrast [1]. It is then smoothed with a Gaussian filter of variance $\sigma^2 = 0.463$:

$$G(x, y) = g(x, y) * I_G(x, y), \quad g(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}} \quad (1)$$

Since DRIVE images have a black background outside the circular field of view (FOV), we additionally fill this region with the mean intensity of the retina before filtering; this step is not described in [1] but was necessary in practice (Section 4).

2.2 Optimized top-hat

Both branches use an optimized top-hat, applied to the complement image $I^c = 1 - I$:

$$T_O = I^c - (I^c \circ S_o) \bullet S_c \quad (2)$$

where S_o, S_c are disk-shaped structuring elements for opening and closing. Radii are (8, 16) for thick vessels and (4, 20) for thin vessels, as in [1]. A second pass with larger radii (32, 86) refines the thick branch after a 2×2 median filter, which removes salt-and-pepper noise introduced by the homomorphic step.

Normalization inside this operator uses the 1st and 99th percentiles of the intensity distribution (restricted to the FOV) instead of the exact minimum/maximum, since a few outlier pixels were otherwise found to compress all useful contrast into a negligible fraction of the dynamic range.

2.3 Homomorphic filtering

Illumination and reflectance are separated in the frequency domain. Taking $I = l \cdot r$, we compute $z = \ln(I)$, apply a Gaussian high-pass filter

$$H(u, v) = 1 - \exp\left(-\frac{D^2(u, v)}{2\sigma^2}\right) \quad (3)$$

to $\mathcal{F}\{z\}$, and undo the logarithm with an exponential after the inverse transform. As in [1], $\sigma = 2$ for thick vessels and $\sigma = 20$ for thin vessels.

2.4 Matched filter (thin branch)

A Gaussian cross-section kernel ($\sigma = 0.8$, 7×7 pixels) is rotated in 26 steps of 7° (from 0° to 182°) and convolved with the image; the maximum response across rotations is retained, following the model of vessel cross-sections used in [4].

2.5 MCET-HHO (thin branch)

The thin branch is binarized using Minimum Cross-Entropy Thresholding with four thresholds, optimized via Harris Hawks Optimization [2]. The objective function is

$$f_{cross}(th) = \sum_{i=1}^L i h(i) \log i - \sum_{k=1}^4 H_k \quad (4)$$

where H_k depends on the mean intensity of each of the five resulting histogram segments. HHO simulates a population of hawks that alternates between exploration and exploitation phases, guided by a time-decaying energy parameter, to minimize f_{cross} . We implemented HHO ourselves (not covered in the course syllabus) with the same hyperparameters as [1]: 250 iterations, 30 hawks. The article does not specify which of the four resulting thresholds separates vessel from background; we use the second-highest, chosen empirically.

2.6 Thick branch binarization and post-processing

The exact binarization rule for the thick branch is not given in [1]; we use the 20th percentile of the refined thick-branch intensities within the FOV. The two branches are merged with a logical OR, small connected components (fewer than 30 pixels) are removed, and a morphological closing (radius 2) reconnects fragmented vessels, in that order — reversing it was found to amplify noise instead of removing it.

3 Differences with respect to the reference article

The article reports its metrics on the DRIVE *test* subset (20 images, two independent observers). We only had ground-truth annotations for the *training* subset (images 21–40, single observer), so all results below are computed on this subset. Both subsets share acquisition conditions, but the numbers are not strictly comparable.

4 Practical difficulties

Loading DRIVE’s LZW-compressed `.tif` images with `plt.imread` or `PIL` silently crashed the Python kernel; the `tifffile` library (with the `imagecodecs` package) solved this. Early pipeline runs also produced masks entirely dominated by the FOV boundary, or classified almost the whole retina as vessel; both traced back to the sharp black-background/retina intensity jump overwhelming the top-hat and homomorphic filters, solved by the background fill and FOV erosion described above.

5 Results

Figure 1 shows a representative example (image 21) comparing the original fundus image, our final segmentation mask, and the manual ground truth. The overall vascular structure is recovered, although some thin peripheral vessels are missed, consistent with the sensitivity values discussed below.

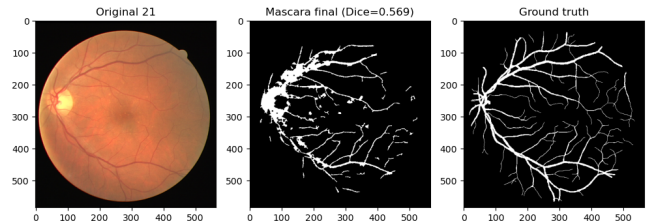


Figure 1: Example segmentation on image 21 of the DRIVE training set. Left: original fundus image. Center: our final segmented mask (Dice = 0.569). Right: manual ground truth.

Table 1 summarizes the metrics on the 20 training images, alongside the values reported in [1] (test set, for reference only).

Table 1: Segmentation performance

	Ours (training)	[1] (test)
Accuracy	0.854 ± 0.047	0.967
Sensitivity	0.466 ± 0.097	0.758
Specificity	0.910 ± 0.054	0.986
Dice	0.451 ± 0.086	—

6 Discussion

Accuracy and specificity are reasonably close to [1]; sensitivity is markedly lower, meaning our pipeline is conservative: false positives are rare, but a fair number of thin vessels are missed, especially in lower-contrast images. This fits with the two undocumented decisions we had to make on our own (the thick-branch threshold, and the choice among MCET-HHO’s four thresholds), which the original authors likely tuned with more care than we could. The original article itself lists low sensitivity and FOV-border artifacts as known weaknesses of the method, and we ran into both, which suggests these are baked into the approach rather than just gaps in our implementation.

Figure 2 shows the final mask obtained for each of the 20 training images. Performance is clearly not uniform: images such as 22, 32 or 33 (Dice around 0.55–0.61) show a well-defined vascular tree, while images such as 26, 34 or 37 (Dice below 0.37) are dominated by a thick, geometric ring of false positives near the FOV border. Since the MCET-HHO thresholds are optimized independently for each image, some runs converge to a solution that is more sensitive to this border artifact than others, which explains most of the standard deviation reported in Table 1.

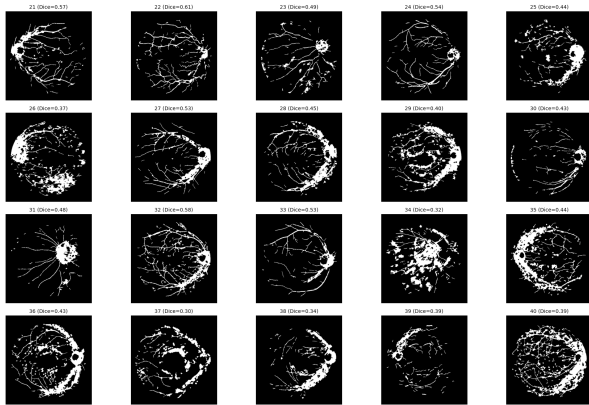


Figure 2: Final segmentation mask obtained for each of the 20 DRIVE training images, with the corresponding Dice coefficient. Images with a pronounced ring near the FOV border (e.g. 26, 34, 37) correspond to the lowest Dice scores.

7 Conclusions

We reproduced the full pipeline of [1], including a from-scratch implementation of the HHO metaheuristic, using only course-covered libraries. The overall structure and, notably, the same qualitative weaknesses (low sensitivity, border artifacts) came through in our version too. The numbers themselves are lower than the article’s, mostly because two binarization steps are left unspecified in the paper and we had to fill them in ourselves, and because we evaluated on a different (though comparable) subset of images.

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